

AL JAHRA SULAIBIYA, Kuwait

WMO: 405870

Lat: 29.267N

Lon: 47.717E

Elev: 54

StdP: 100.68

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.3	6.0	-11.5	1.4	13.7	-9.5	1.7	16.2	10.5	14.9	9.6	15.7	3.6	290	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	49.0	20.7	48.0	20.6	47.0	20.4	27.3	34.7	25.4	35.6	23.5	35.3	5.3	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.5	33.4	22.5	17.3	30.8	20.1	14.9	29.4	87.0	33.6	77.9	36.0	70.2	34.9	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	10.0	8.9	DB	1.8	51.1	1.9	1.0	0.5	51.8	-0.7	52.4	-1.7	53.0	-3.1	53.7	
			WB	-1.2	29.0	2.1	0.9	-2.7	29.6	-4.0	30.2	-5.2	30.7	-6.7	31.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	13.8	16.6	21.6	27.0	33.9	38.4	40.0	39.7	35.8	30.0	20.8	15.5
	DBStd	9.86	3.21	3.39	3.61	3.28	2.94	2.17	1.75	1.61	2.08	2.90	3.47	2.95
	HDD10.0	8	7	1	0	0	0	0	0	0	0	0	0	1
	HDD18.3	325	142	66	10	0	0	0	0	0	0	0	14	93
	CDD10.0	6512	126	186	359	509	740	853	929	920	774	621	324	172
	CDD18.3	3787	3	17	110	259	482	603	671	662	524	363	88	7
	CDH23.3	61986	18	122	1041	3022	7681	10933	12427	12121	8944	4874	739	64
Wind	WSAvg	4.1	3.4	3.9	3.9	3.9	4.2	5.5	5.4	4.5	4.0	3.4	3.3	3.4
	PrecAvg	111	23	11	16	10	3	0	0	0	0	3	18	22
	PrecMax	209	95	69	64	45	25	1	1	1	11	38	115	68
	PrecMin	36	1	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	45	23	15	17	11	6	0	0	0	2	7	27	22
	DB	25.5	30.0	38.7	41.2	46.9	49.8	50.4	49.8	47.4	42.7	35.1	27.5	
	MCWB	13.6	14.7	16.2	18.3	20.3	20.4	21.0	21.1	20.9	19.0	16.9	13.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	23.0	26.4	34.2	38.1	44.4	48.4	49.1	48.8	45.5	40.8	32.2	25.1	
	2% MCWB	13.3	13.7	15.7	17.8	19.5	20.2	20.8	20.9	20.4	19.1	16.5	14.0	
	5% DB	21.2	24.5	31.0	35.9	42.7	47.0	48.0	47.8	44.2	39.1	29.7	23.1	
	5% MCWB	12.9	13.2	14.9	16.9	19.1	19.9	20.6	20.8	20.0	18.8	15.7	13.9	
	10% DB	19.6	22.9	28.4	34.0	41.0	45.7	46.9	46.7	43.1	37.5	27.5	21.4	
	10% MCWB	12.2	12.7	14.4	16.6	18.8	19.7	20.4	20.7	19.7	18.2	15.2	13.2	
	WB	17.5	17.8	19.8	21.1	23.2	24.9	27.7	28.9	28.7	26.6	22.7	19.0	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	20.3	22.0	25.7	28.9	33.9	39.2	34.5	35.5	33.2	31.5	26.1	21.5	
	2% WB	16.3	16.7	18.8	20.1	21.7	21.0	24.3	27.7	26.9	24.7	20.6	17.5	
	2% MCDB	19.4	21.3	23.9	28.3	34.5	46.1	38.8	35.2	34.0	29.8	24.9	21.3	
	5% WB	14.8	15.6	17.8	19.4	20.8	20.4	22.0	25.7	24.5	23.4	18.7	16.0	
	5% MCDB	18.7	20.6	24.0	28.6	34.1	45.6	43.7	37.0	36.7	29.3	23.2	19.3	
	10% WB	13.4	14.5	16.5	18.6	20.1	19.9	21.0	23.1	21.8	22.0	17.7	14.4	
	10% MCDB	17.7	20.1	25.0	29.0	36.1	45.0	45.8	40.1	37.3	28.1	22.8	19.3	
Mean Daily Temperature Range	MDBR	11.0	11.7	13.0	12.8	13.1	14.1	14.2	14.9	15.1	13.8	11.8	11.8	
	5% DB	12.7	13.5	17.1	15.3	14.9	15.6	15.0	15.9	16.0	15.7	14.8	13.9	
	MCWBR	5.5	5.5	6.1	4.9	5.0	5.0	4.7	5.2	5.5	5.9	6.0	6.0	
	5% WB	10.4	10.7	11.0	11.8	13.0	15.0	15.3	14.7	14.4	12.3	9.3	9.5	
	MCWBR	5.5	5.5	5.5	4.4	4.6	5.1	6.1	6.9	7.3	5.8	4.7	5.3	
Clear-Sky Solar Irradiance	taub	0.505	0.554	0.610	0.699	0.702	0.658	0.673	0.624	0.572	0.574	0.546	0.508	
	taud	1.794	1.698	1.583	1.481	1.481	1.525	1.525	1.613	1.700	1.700	1.746	1.801	
	Ebn at Noon	695	702	696	655	657	684	671	699	717	675	650	666	
	Edh at Noon	184	221	263	301	302	287	286	260	231	217	192	175	
All-Sky Solar Radiation	RadAvg	3.51	4.37	5.50	6.01	6.91	7.73	7.42	7.05	6.27	4.97	3.65	3.23	
	RadStd	0.25	0.27	0.23	0.30	0.31	0.26	0.20	0.15	0.20	0.21	0.28	0.29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.39	N/A	N/A	+0.59	N/A	N/A	N/A	N/A	+129	+136

Nomenclature: See separate page